


- Per-unit Tax, is a tax on the quantity being sold, you pay t per unit bought.

• If the price is p and you consume q

$$\rightarrow \text{Total you pay is } pq + t \cdot q = (p + t)q$$

- Ad-Valorem Tax: tax on value being bought.

(VAT)

If you buy q units at price p ,

Total value: $p \cdot q \leftarrow$ this is the tax base

$$\text{Tax } \tau: \tau p \cdot q$$

$$\text{Total expenditure } p \cdot q + \tau p \cdot q = p \cdot q (1 + \tau)$$

Summary of Tax Incidence

- The more inelastic side pays more of the tax burden
 - Perfectly Inelastic Demand $\rightarrow$ Buyers pay 100% of tax burden
 - Perfectly Inelastic Supply $\rightarrow$ Sellers pay 100% of tax burden
- The more elastic side pays less of the tax burden
 - Perfectly Elastic Demand $\rightarrow$ Sellers pay 100% of tax burden
 - Perfectly Elastic Supply $\rightarrow$ Buyers pay 100% of tax burden

Tax Revenue for a per unit tax is $T \times Q$

Tax Incidence:

Buyers: Price Paid Consumers - $P_{\text{equilibrium}}$

$$\text{Share: } \frac{\text{Price Paid Consumers} - P_{\text{equilibrium}}}{\text{Tax}}$$

Sellers: $P_{\text{equilibrium}}$ - Price Received by Sellers

$$\text{Share: } \frac{P_{\text{equilibrium}} - \text{Price Received by Sellers}}{\text{Tax}}$$

Subsidies:

- per-unit subsidies : $S \rightarrow (p - S) \cdot q$

↳ applied per unit

- ad-valorem subsidies : $\sigma \rightarrow (p \cdot q)(1 - \sigma) = p(1 - \sigma)q$

↳ applied to total value